AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 28 · WORKBOOK

Prompt Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected prompt-comparison record.

Explain the contract, trace paired outcomes, audit the limit and design a reproducible follow-up.

Keep one habit: Retain total and slice counts, gains/losses, selection priorities and unmeasured conditions.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Predictive vs Generative

Text as Pieces

Nearest Neighbor Reasoning

W1 · Explain a controlled prompt comparison

> Name what changes and what must remain fixed.

Your evidence record

- Six-case W2 record: P passes1,2,3,4; Q passes1,2,3,5,6. Missing-approval IDs4,6. Separate seven-case picture: same pass IDs but total7 and missing-source IDs4,7. These are different datasets. Do not substitute the picture denominator or slice into W2. All outcomes are invented development records, not independently measured final performance.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace six paired cases

> Compute totals and the two-case missing-approval slice.

Your evidence record

- Six-case W2 record: P passes1,2,3,4; Q passes1,2,3,5,6. Missing-approval IDs4,6. Separate seven-case picture: same pass IDs but total7 and missing-source IDs4,7. These are different datasets. Do not substitute the picture denominator or slice into W2. All outcomes are invented development records, not independently measured final performance.

- On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Separate development selection from final evidence

> A reused selection set is not an untouched final test.

Your evidence record

- Six-case W2 record: P passes1,2,3,4; Q passes1,2,3,5,6. Missing-approval IDs4,6. Separate seven-case picture: same pass IDs but total7 and missing-source IDs4,7. These are different datasets. Do not substitute the picture denominator or slice into W2. All outcomes are invented development records, not independently measured final performance.

- A report omits this qualification: **Prompt** selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan a revealing follow-up

> Specify cases, criteria and a repeated-run plan before collecting outputs.

Your evidence record

- Six-case W2 record: P passes1,2,3,4; Q passes1,2,3,5,6. Missing-approval IDs4,6. Separate seven-case picture: same pass IDs but total7 and missing-source IDs4,7. These are different datasets. Do not substitute the picture denominator or slice into W2. All outcomes are invented development records, not independently measured final performance.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Compare a new paired record independently

> Keep one first attempt before hints or feedback.

Read and explain independently

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Test generated book explanations with a fixed **rubric** and prohibit invented details.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Controlled **prompt** experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Controlled **prompt** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Prompt A is tested on easy requests and B on requests with missing data. B scores lower. Design a fair comparison and specify what to do when **output**s vary across runs.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Prompt B adds under 100 pages and explain the match to **Prompt** A. Both use a catalogue.

Design a fair comparison.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Prompt A and B are tested on different questions. B scores higher. Can wording alone explain the difference? Design a **controlled comparison**.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 28 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Prompt A is tested on easy requests and B on requests with missing data. B scores lower. Design a fair comparison and specify what to do when **output**s vary across runs.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional book-help project tests two **prompt** versions on seven shared development cases.

P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

AI ACADEMY / AI-2/28

Paired overall and missing-source passes

	Overall passes /7	Missing-source passes /2
P	4	1
Q	5	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/28

Prompt Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired setup

Use the same seven cases and pass criteria for both prompts.



2. Overall rates

P=4/7; Q=5/7.



3. Missing-source slice

P passes 1/2; Q passes 0/2 in cases 4 and 7.



4. Decide explicitly

Q improves overall count but worsens missing-source behavior; use the declared priorities.

CHANGE ONE THING

Make missing-source correctness a required selection gate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Name the case P passes and Q fails, and explain why it matters.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-help project tests two **prompt** versions on seven shared development cases.
- Which labelled step matches this detail? Q improves overall count but worsens missing-source behavior; use the declared priorities.
- 1. Decide explicitly
- 2. Paired setup
- 3. Missing-source slice
- 4. Overall rates
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Decide explicitly Q improves overall count but worsens missing-source behavior; use the declared priorities.

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

Changed-case question

- Changed-case practice: Make missing-source correctness a required selection gate. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Neither **prompt** clears a 2/2 gate on these records.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Neither **prompt** clears a 2/2 gate on these records. P fails 7 and Q fails both 4 and 7; both need revision.

Repair and save your work

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

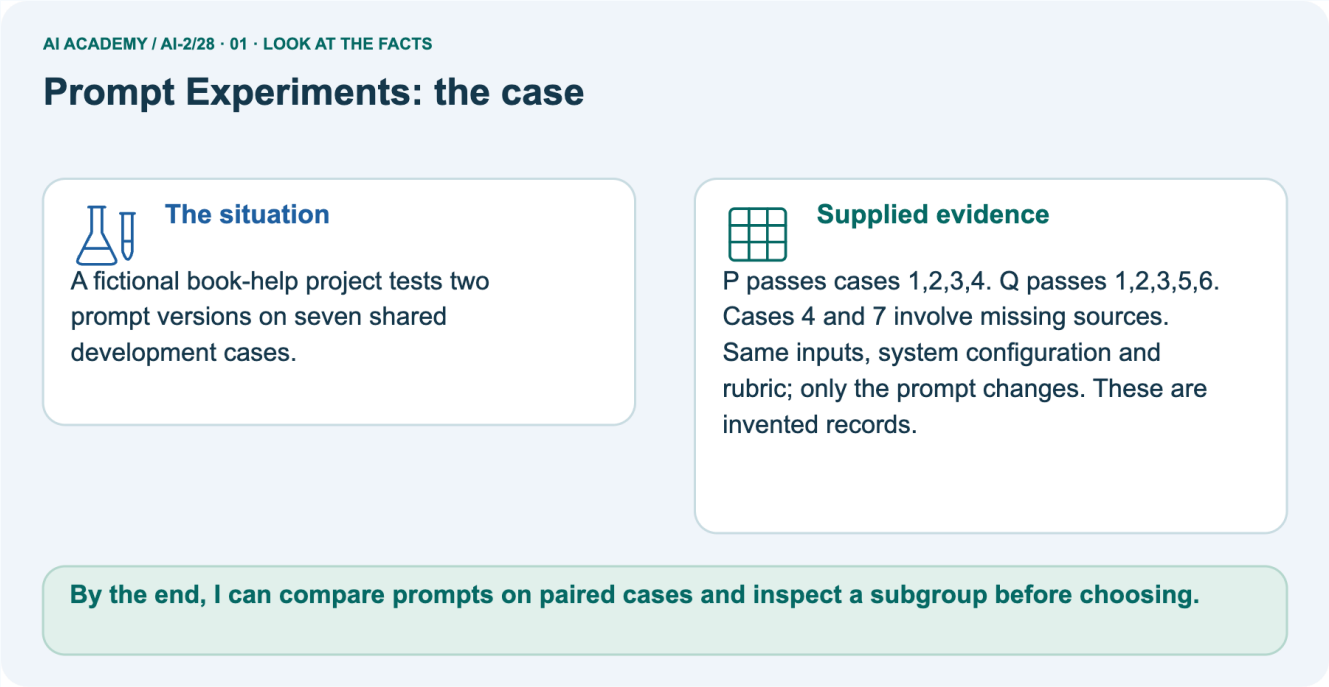
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Prompt Experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

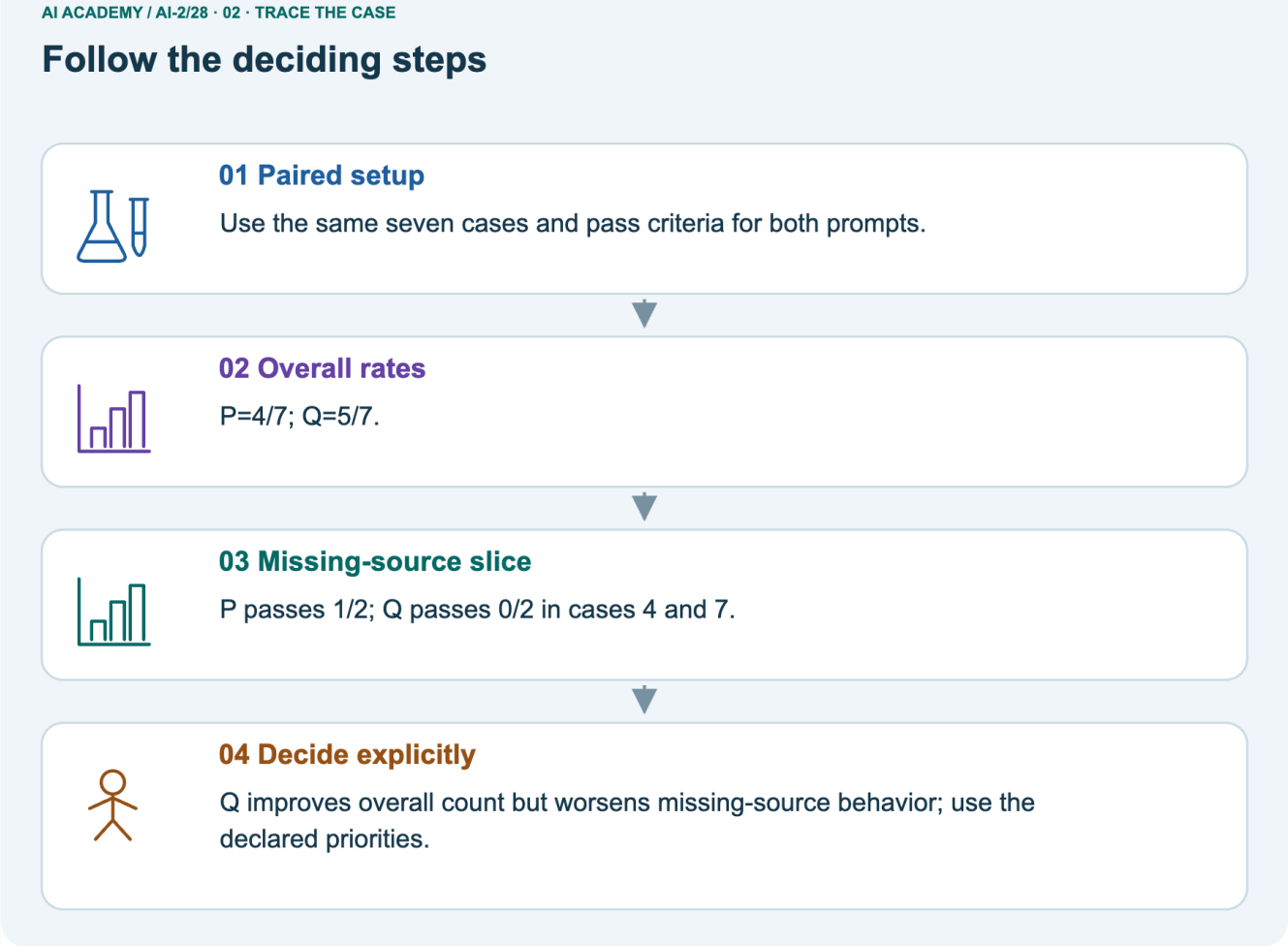
A fictional book-help project tests two **prompt** versions on seven shared development cases. Supplied facts: P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



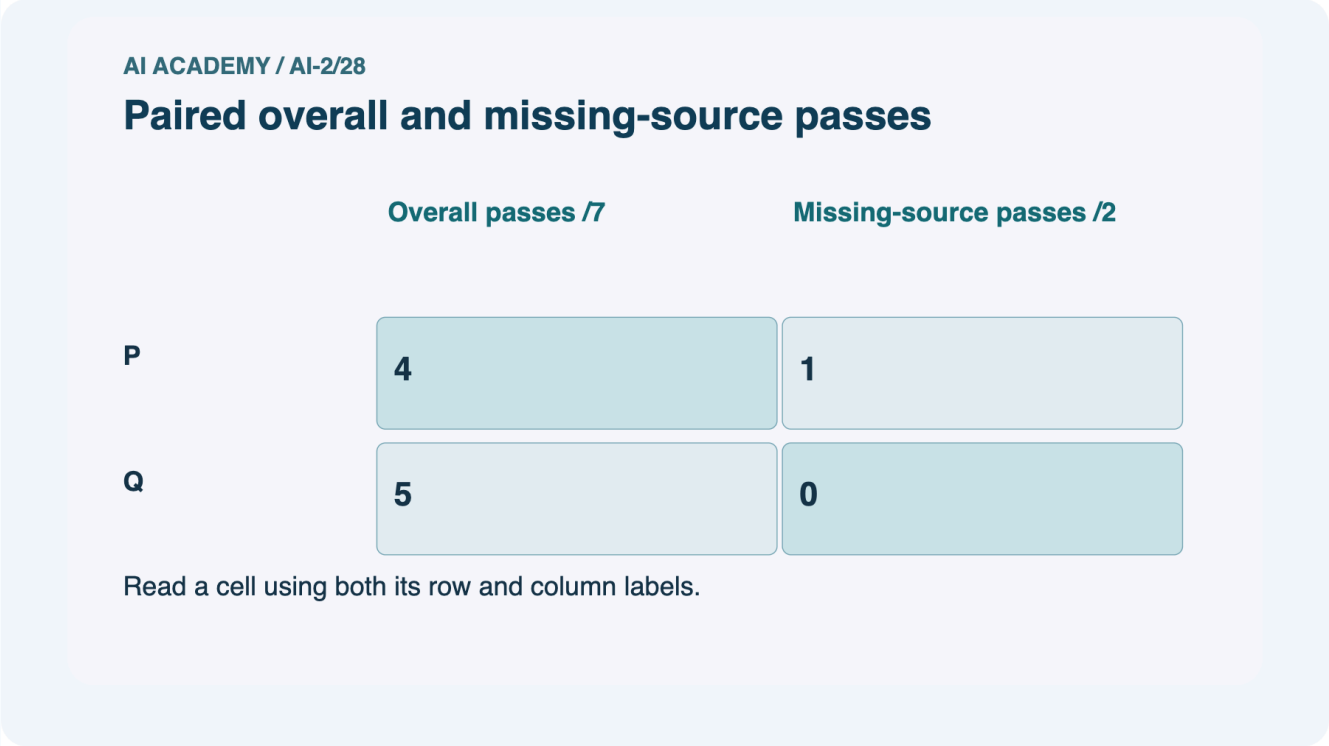
Follow the labelled picture with the matching explanation. Open the vector diagram

Paired setup: Use the same seven cases and pass criteria for both **prompt**s. → Overall rates: P=4/7; Q=5/7. → Missing-source slice: P passes 1/2; Q passes 0/2 in cases 4 and 7. → Decide explicitly: Q improves overall count but worsens missing-source behavior; use the declared priorities.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Paired overall and missing-source passes



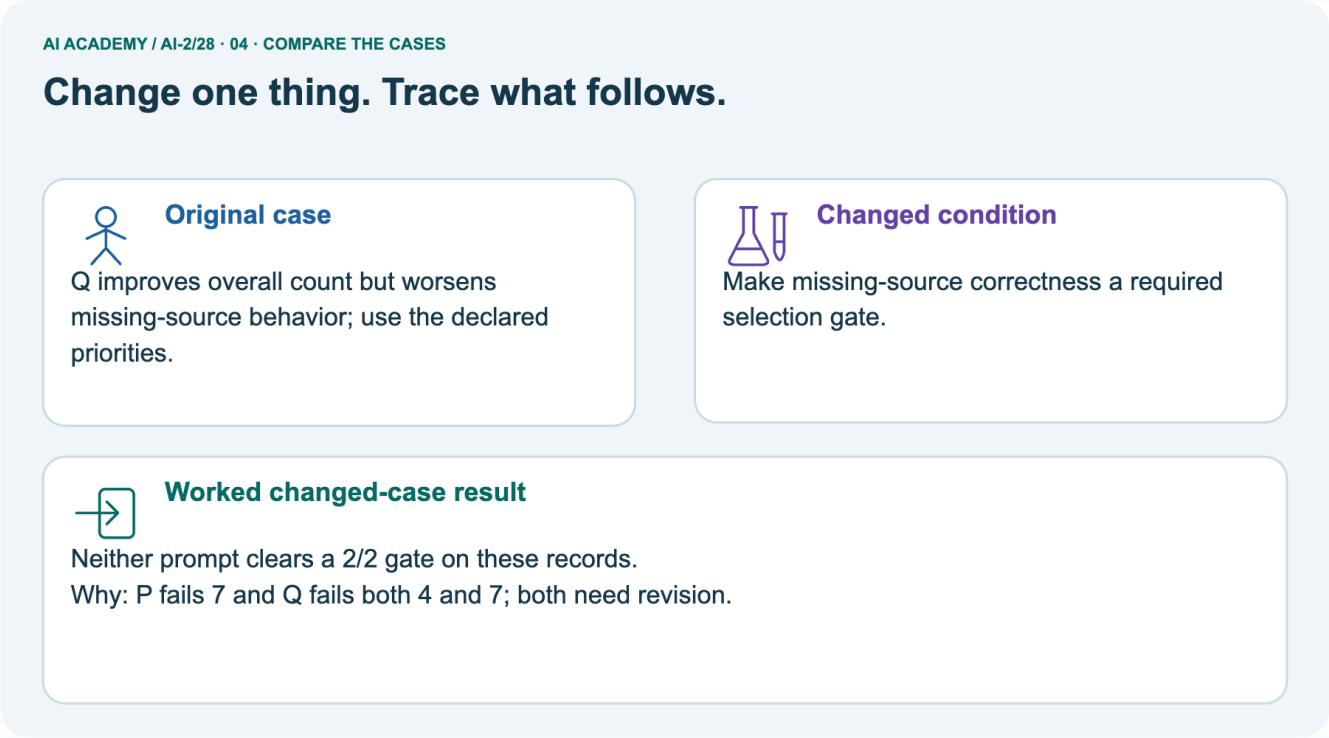
Follow the labelled picture with the matching explanation. Open the vector diagram

Paired overall and missing-source passes. P: Overall passes /7 = 4, Missing-source passes /2 = 1. Q: Overall passes /7 = 5, Missing-source passes /2 = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

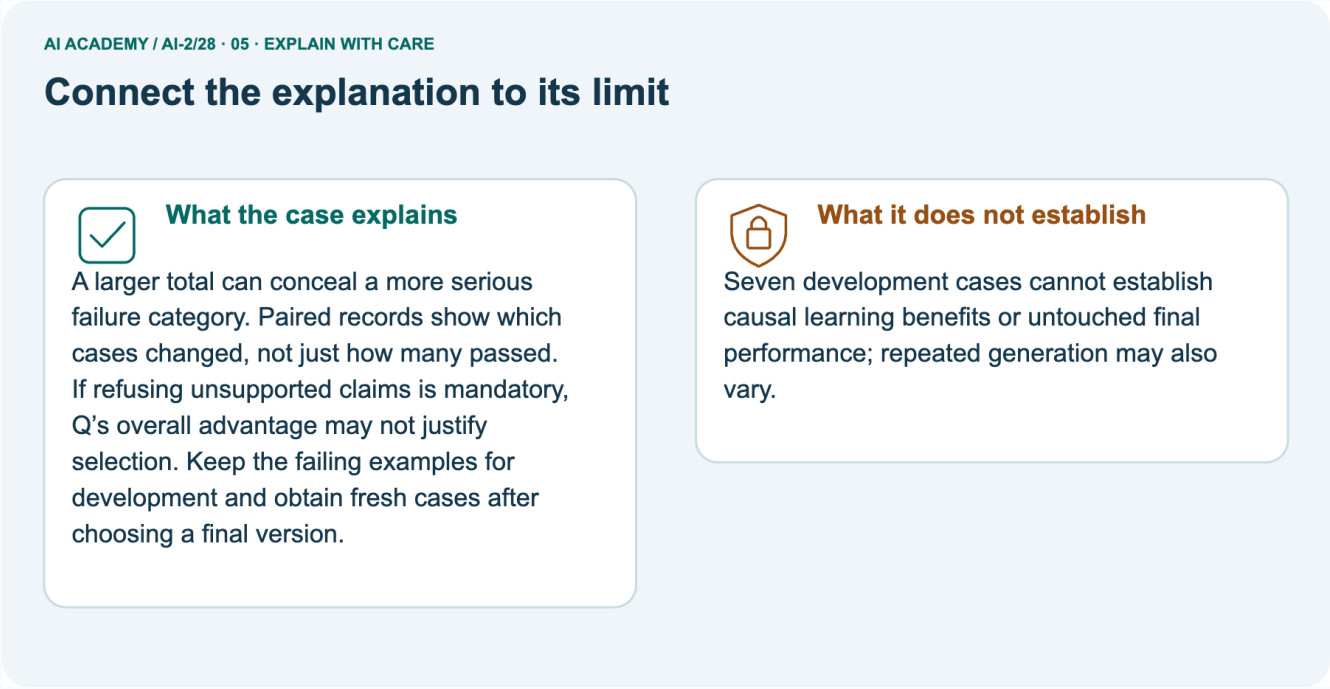
Original case: Q improves overall count but worsens missing-source behavior; use the declared priorities. Changed condition: Make missing-source correctness a required selection gate.

Expected result: Neither **prompt** clears a 2/2 gate on these records. Reason: P fails 7 and Q fails both 4 and 7; both need revision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

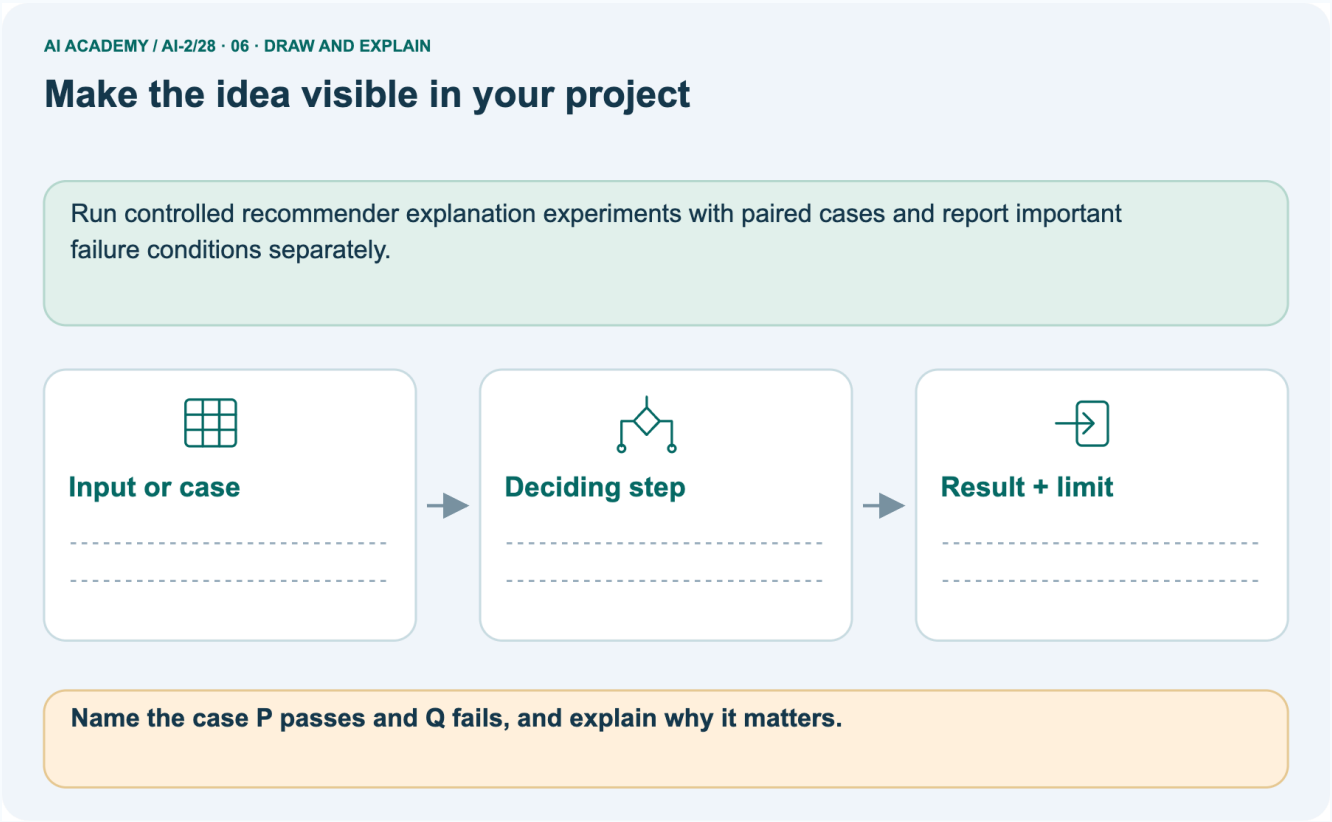
Keep the failing examples for development and obtain fresh cases after choosing a final version.

Limit: Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately. Draw your **input**, deciding step, result and limit.

Supported question: Name the case P passes and Q fails, and explain why it matters.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Prompt Experiments: the case

02. Follow the deciding steps

03. Paired overall and missing-source passes

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.